AI-Based Student Performance Prediction System

Abstract

This project predicts student academic performance using machine learning techniques based on study hours, attendance, assignments, and previous marks.

Objective

To predict student performance and identify students who need academic support using AI and machine learning.

Technologies Used

Python, Pandas, NumPy, Scikit-learn, Matplotlib, Jupyter Notebook

Workflow

1. Data Collection
2. Data Preprocessing
3. Feature Selection
4. Model Training
5. Prediction
6. Evaluation

Advantages

Easy prediction of performance, supports teachers, saves time, and improves monitoring.

Applications

Schools, colleges, online learning platforms, and educational analytics systems.

Conclusion

The project successfully predicts student academic performance using machine learning algorithms.

Dataset Sample

StudyHours	Attendance	Assignments	PreviousMarks	FinalResult
2	60	3	50	55
3	70	4	55	60
5	80	6	65	70

6	90	7	70	75

Python Code

```
import pandas as pd
from sklearn.linear_model import LinearRegression

# Load dataset
df = pd.read_csv('dataset.csv')

# Features and target
X = df[['StudyHours', 'Attendance', 'Assignments', 'PreviousMarks']]
y = df['FinalResult']

# Train model
model = LinearRegression()
model.fit(X, y)

print('Model Trained Successfully')
```

Submission Checklist

- LinkedIn Profile Link
- GitHub Repository named upskillCampus
- Upload final code and report
- Update Table of Contents
- Submit GitHub link in Google Form